

SIOC 251: Radiative Transfer

Homework 3: Monte Carlo Radiative Transfer with Scattering and Clouds

1 Overview

In this assignment you will extend your Monte Carlo radiative transfer model to include scattering, multiple scattering, non-isotropic phase functions, and clouds. The goal is to understand how scattering controls the partitioning of solar radiation between reflection to space, absorption in the atmosphere, and transmission to the surface.

You will:

- add scattering to your Monte Carlo radiative transfer model
- implement multiple scattering
- use the Henyey-Greenstein phase function
- simulate the radiative effects of a cloud layer
- investigate how cloud microphysics controls cloud optical depth and cloud radiative effect

For this exercise, unless specifically requested, assume that optical properties are not a function of wavelength.

2 Submission Timeline

This assignment will be completed in three stages:

- **May 14:** Part I (Isotropic Scattering)
- **May 21:** Part II (Non-Isotropic Scattering)
- **May 28:** Part III (Cloud Radiative Effects)

Each stage must be submitted as a working notebook in it's own repository structure with:

- a functioning model
- verification tests
- answers to the questions for that section

3 Repository Structure and Submission

```
sioc251_radiative_transfer
|
|-- HW1_monte_carlo_extinction
|
|-- HW2_longwave_model
|
|-- HW3_monte_carlo_scattering_PartI
    |
    |-- scattering_model.ipynb
    |-- (optional additional notebooks)
    |
    |-- figures
        |
        |-- figure1.png
```

- Final submission must be a **Python notebook**
- Save figures in **figures/**
- Push to your GitHub repository
- Include a short AI usage statement at the end of the notebook

4 AI Use Policy

Students are encouraged to use AI tools (e.g., ChatGPT, Claude, Gemini) to assist with coding, debugging, plotting, and documentation. However, students remain fully responsible for the scientific correctness and physical interpretation of their work. AI output should be treated like information from other sources: it must be understood, verified, and appropriately documented. At the end of your notebook, include a short statement describing:

- which AI tools you used, if any
- how you used them
- what parts of the work you checked independently
- at least one example of something you verified rather than simply accepting from AI output

5 Part I: Isotropic Scattering

Start with your Monte Carlo model from Homework 1. In that model, photons were either absorbed in the atmosphere or transmitted through the atmosphere and absorbed at the surface. We will now allow photons to scatter.

5.1 Model Construction

- Modify your model so that, after each photon path length is calculated, the photon can either be absorbed or scattered. The probability that an interaction is a scattering event is represented by the single scatter albedo, $\tilde{\omega}$. If ζ is drawn from a uniform random distribution between 0 and 1, then a scattering event occurs when $\zeta < \tilde{\omega}$ and absorption occurs otherwise.
- For isotropic scattering, the probability distribution for scattering angle is uniform in $\cos \Theta$:

$$P(\cos \Theta) = 1$$

The cumulative distribution function is

$$C(\cos \Theta) = \frac{1}{2} \int_{-1}^{\cos \Theta} P(x) dx$$

where x is a dummy variable. Solve this expression for $\cos \Theta$, assuming that $C(\cos \Theta)$ can be represented by a random number drawn from a uniform distribution between 0 and 1.

- Choose a scattering angle in the azimuthal direction, Φ , by drawing another random number from a uniform distribution between 0 and 2π .
- Convert the scattering angles Θ and Φ , which are defined relative to the incoming photon trajectory, into your model coordinates (see Appendix).
- After a photon scatters you need to repeat the process of picking a new optical path length to travel and determine if it is absorbed, scatters again (in which case you have to repeat the above), or exits out the top or bottom of the atmosphere.

5.2 Verification

Demonstrate that your model behaves correctly via two model verification tests that examine model behavior for limiting cases.

5.3 Questions

Once your single-scattering model is working, use your model to answer the following questions

- For $\tilde{\omega}$ increasing from $0 \rightarrow 1$, how does the the surface direct and diffuse transmittance, and the albedo, change?
- For $\tau \ll 1$, what is the relationship between diffuse radiation at the surface and $\tilde{\omega}$, τ , and μ_0 , and how does this compare to an approximate analytical solution to the RTE?
- Prove that the sky is blue and sunsets are red.
- Come up with your own question & hypothesis that can be answered by the model, run the simulations, interpret the results, and answer your question.

6 Part II: Non-Isotropic Scattering

We will now increase the complexity of our model by incorporating a phase function into our scattering routine.

6.1 Model Construction

- a. Replace isotropic scattering in your model with the Henyey-Greenstein phase function:

$$P_{HG}(\cos \Theta) = \frac{1 - g^2}{(1 + g^2 - 2g \cos \Theta)^{3/2}}$$

where g is the asymmetry parameter. To do so you will need to calculate $C(\cos \Theta)$ for this case, which will then allow you to estimate the scattering angle from a uniform distribution.

- b. Follow the remaining instructions for Part I.

6.2 Verification

- a. Further demonstrate that your model behaves correctly via at least one additional model verification test that examines model behavior for a limiting case.

6.3 Questions

Use your model to answer the following questions

- a. Using theory alone, show that for $\tau \ll 1$ and $g = 0$, the albedo is $\alpha \approx \tau \tilde{\omega}/2$. You can assume that the solar zenith angle is 0° , so that $\mu_0 = 1$.
- b. Use your model to demonstrate the above relation is true.
- c. Use your model to describe the functional relationship between α , $\tilde{\omega}$, and g for $\tau \ll 1$ and $\mu_0 = 1$. Define

$$f(g) = \frac{\alpha}{\tau \tilde{\omega}}.$$

You are welcome to choose one representative value of $\tilde{\omega}$ and vary only g .

- d. The backscatter function, $B(g)$, is the fraction of scattered light directed into the backward hemisphere. Let $\mu = \cos \Theta$, where Θ is the scattering angle. Then

$$B(g) = \frac{1}{2} \int_{-1}^0 P_{HG}(\mu, g) d\mu.$$

For the conditions you set in part c., derive the analytical solution for $B(g)$ and compare it with the numerical results obtained from your model.

- e. Come up with your own question & hypothesis that can be answered by the model, run the simulations, interpret the results, and answer your question.

7 Part III: Cloud Radiative Effects

We are going to add a low marine cloud to your atmosphere to model aerosol direct and indirect effects.

7.1 Model Construction

- a. Construct your atmosphere as having three layers: 1) clear atmosphere above the cloud, 2) cloud layer, 3) clear atmosphere below the cloud
- b. Assume that the cloud has $\tilde{\omega} = 1$, $g = 0.8$
- c. estimate the cloud optical depth using these physical and microphysical properties:
 - Cloud-top height: 1 km
 - Cloud thickness: 250 m
 - R_{eff} : $11 \mu\text{m}$
 - LWC: 0.30 g m^{-3}
- d. Assume that the surface is perfectly absorbing.
- e. Estimate a reasonable clear-sky optical depth above and below the cloud. One simple approach is to assume that optical depth is proportional to pressure. The clear-sky atmosphere should only cause extinction via isotropic scattering.
- f. assume that all optical properties are constant across all visible wavelengths.

7.2 Verification

- a. Demonstrate that your model behaves correctly via at least two model verification tests that examine the model behavior for limiting cases.

7.3 Questions

Use your model to answer the following questions

- a. What is the SW cloud radiative effect (CRE) at the top of the atmosphere (W m^{-2})?
- b. Vary R_{eff} from half to double the initial value while holding LWC and the cloud geometric dimensions constant. Plot the CRE as a function of R_{eff} and calculate the slope of this line (units of $\text{W m}^{-2} \mu\text{m}^{-1}$). This is the first aerosol indirect effect.
- c. Come up with your own question & hypothesis that can be answered by the model, run the simulations, interpret the results, and answer your question.

A Appendix: Scattering Geometry and Coordinate Rotation

After a photon scatters, the scattering angles Θ and Φ are defined relative to the incoming photon direction, not relative to the model coordinate system. Therefore, the scattered direction must be rotated from the local photon coordinate system back into the model coordinates.

This appendix walks through that rotation using a specific example. The goal is to make sure that the scattered photon continues traveling in the correct physical direction.

A.1 Incoming Photon Direction

Suppose the incoming photon has zenith angle θ_0 and azimuth angle ϕ_0 . In model coordinates, define the photon propagation vector as

$$\mathbf{k}_0 = \begin{pmatrix} \cos \phi_0 \sin \theta_0 \\ \sin \phi_0 \sin \theta_0 \\ -\cos \theta_0 \end{pmatrix}$$

For example, if

$$\theta_0 = 50^\circ, \quad \phi_0 = 37^\circ,$$

then

$$\mathbf{k}_0 = \begin{pmatrix} 0.387 \\ 0.461 \\ -0.799 \end{pmatrix}$$

The negative vertical component indicates that the photon is initially traveling downward.

A.2 Define the Photon Coordinate System

We now define a local photon coordinate system. The local z_p axis is aligned with the incoming photon direction:

$$\mathbf{z}_p = \mathbf{k}_0$$

The local x_p axis is chosen to be horizontal in model coordinates and perpendicular to $\mathbf{z}_p$:

$$\mathbf{x}_p = \frac{1}{\sqrt{k_x^2 + k_y^2}} \begin{pmatrix} -k_y \\ k_x \\ 0 \end{pmatrix}$$

Using the example above,

$$\mathbf{x}_p = \begin{pmatrix} -0.766 \\ 0.643 \\ 0 \end{pmatrix}$$

Note: If the photon is traveling exactly vertically, then $k_x = k_y = 0$ and this expression is undefined. In that special case set

$$\mathbf{x}_p = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

Next, the local y_p axis is

$$\mathbf{y}_p = \mathbf{z}_p \times \mathbf{x}_p$$

which for our example is,

$$\mathbf{y}_p = \begin{pmatrix} 0.513 \\ 0.612 \\ 0.602 \end{pmatrix}$$

The three vectors $\mathbf{x}_p$, $\mathbf{y}_p$, and $\mathbf{z}_p$ define the coordinate system attached to the incoming photon.

A.3 Scattered Direction in Photon Coordinates

Now choose the scattering angles Θ and Φ . These angles are relative to the incoming photon direction.

The scattered photon direction in photon coordinates is

$$\mathbf{k}_p = \begin{pmatrix} \sin \Theta \cos \Phi \\ \sin \Theta \sin \Phi \\ \cos \Theta \end{pmatrix}$$

For example, if

$$\Theta = 30^\circ, \quad \Phi = 135^\circ,$$

then

$$\mathbf{k}_p = \begin{pmatrix} -0.354 \\ 0.354 \\ 0.866 \end{pmatrix}$$

A.4 Rotate Back into Model Coordinates

Construct the rotation matrix whose columns are the photon-coordinate basis vectors expressed in model coordinates:

$$\mathbf{A} = \begin{pmatrix} | & | & | \\ \mathbf{x}_p & \mathbf{y}_p & \mathbf{z}_p \\ | & | & | \end{pmatrix}$$

For the example above,

$$\mathbf{A} = \begin{pmatrix} -0.766 & 0.513 & 0.387 \\ 0.643 & 0.612 & 0.461 \\ 0 & 0.602 & -0.799 \end{pmatrix}$$

The scattered photon direction in model coordinates is

$$\mathbf{k}_{new} = \mathbf{A}\mathbf{k}_p$$

or equivalently,

$$\mathbf{k}_{new} = k_{p,x}\mathbf{x}_p + k_{p,y}\mathbf{y}_p + k_{p,z}\mathbf{z}_p$$

Using the numerical example,

$$\mathbf{k}_{new} = \begin{pmatrix} 0.788 \\ 0.388 \\ -0.479 \end{pmatrix}$$

This is the photon propagation vector after scattering, expressed in the original model coordinates. Next, normalize the rotated vector to avoid numerical drift:

$$\mathbf{k}_{new} \leftarrow \frac{\mathbf{k}_{new}}{|\mathbf{k}_{new}|}$$

Finally, convert the new propagation vector back into model polar angles. Since

$$\mathbf{k}_{new} = \begin{pmatrix} \cos \phi_{new} \sin \theta_{new} \\ \sin \phi_{new} \sin \theta_{new} \\ -\cos \theta_{new} \end{pmatrix}$$

we have

$$\theta_{new} = \cos^{-1}(-k_{new,z}) = \cos^{-1}(0.479) = 61.4^\circ$$

and

$$\phi_{new} = \tan^{-1} \left(\frac{k_{new,y}}{k_{new,x}} \right) = \tan^{-1} \left(\frac{0.388}{0.787} \right) = 26.2^\circ$$

Thus, after scattering, the photon is traveling with

$$\theta_{new} = 61.4^\circ, \quad \phi_{new} = 26.2^\circ$$

in model coordinates.